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23 APRIL 1956
SUPERSEDING
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MILITARY SPECIFICATION

WAX, IMPREGNATING, WATERPROOFING, FOR LAMINATED PAPER TUBES FOR SMALL ARMS AMMUNITION

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers one grade of water-proofing wax for impregnating laminated paper tubes for small arms ammunition by the hot-dipping method. The waterproofing wax is a high-melting-point hydrocarbon, chemically inert, non-hygroscopic wax containing a compound that will not bleed or block.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

SS-R-406 — Road and Paving-Materials; Methods of Sampling and Testing.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 791 — Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The ingredients used in the preparation of the wax shall be of the best commercial quality and shall be such as to produce a wax meeting the requirements of this specification.

3.2 Softening point (dropping point). The softening point of the wax shall be not less than 165°F., when tested as specified in 4.4.1.

3.3 Viscosity. The viscosity of the wax shall be 60 to 105 seconds Saybolt at 210°F., when tested as specified in 4.4.1.

3.4 Flash point. The flash point of the wax shall be not less than 425°F., when tested as specified in 4.4.1.

3.5 Oil content. The oil content of the wax shall not exceed 6 percent, when determined as specified in 4.4.1.

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3.6 Gravity. The gravity of the wax shall be not less than 30 degrees API at 60°F., when determined as specified in 4.4.1.

3.7 Complete liquefaction temperature. The wax shall be completely liquefied when maintained at a temperature of 185°F. for 1 hour, as specified in 4.4.2.

3.8 Stability. The wax shall show no evidence of sludging or tendency to separate after being heated for 72 hours at 210°F., as specified in 4.4.3.

3.9 Corrosive effect. Steel, brass, aluminum and copper shall show no evidence of discoloration or corrosion by the wax, when tested as specified in 4.4.4.

3.10 Needle penetration. The needle penetration of the wax shall be not greater than 85, when tested as specified in 4.4.5.

3.11 Acid number. The acid number of the wax shall not exceed 1.0, when determined as specified in 4.4.6.

3.12 Fungistatic agent. The wax shall be made fungistatic by the addition of a suitable solubilized copper-8-quinolinolate formulation. The addition of the formulation shall be such that the wax contains a concentration of 0.09 to 0.11 percent of solubilized copper-8-quinolinolate.

3.12.1 *Fungistatic effectiveness.* After being treated as specified in 3.12, the wax shall inhibit the growth of fungi when tested as specified in 4.4.7.

3.13 Workmanship. The wax shall be a homogeneously prepared mixture and shall remain stable during use. It shall give off no toxic vapors nor be corrosive to the metals of the dipping equipment, and of the cartridge components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of 20,000 pounds, maximum, of wax submitted at one time for inspection.

4.2 Sampling. The inspection sample shall be obtained by removing equal quantities of wax from each of not less than 1 percent of the containers in the lot, selected so as to be representative of the lot. The individual samples shall be placed together in a suitable heat-resistant container, heated at a temperature not exceeding 210°F., stirring constantly until completely melted. The stirring shall be continued until a homogeneously blended mixture is obtained, and then cast in a clean mold. The blended sample shall be used as the source of all the test samples used in conducting the tests specified hereinafter.

4.3 Inspection. Unless otherwise specified, inspection shall be at the point of delivery.

4.4 Test procedures.

4.4.1 The following tests shall be conducted in accordance with Federal Standard No. 791:

Test	Method
Softening point	1401.2
Viscosity	304.7
Flash point	1103.5
Oil content	5431.2
Gravity	401.4

4.4.2 *Complete liquefaction temperature.* A 200-gram sample of the wax shall be placed in a 400-ml low form, heat-resistant glass beaker and placed in a water bath maintained at a temperature of 185° ± 2°F. for one hour, without stirring. Upon liquefaction of the wax, a thermometer shall be placed so that the bulb is located in the center of the wax. The sample shall be examined visually at the end of the heating period to determine the extent of liquefaction of the wax.

4.4.3 *Stability.* A 200-gram sample of the wax shall be placed in a 400-ml heat-resistant glass beaker and heated, without stirring, for 72 hours in an oven maintained at 210° ± 5°F. At the end of the heating period the wax shall be removed and examined visually for

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evidence of separation of components and sludge. (Examination may be facilitated by pouring the molten wax into a second beaker and examining the bottom and sides of the first beaker for separated material.)

4.4.4 *Corrosive effect.* Polished strips of steel, brass, aluminum, and copper, each measuring approximately ½ by 3 by ⅛ inches, shall be placed separately in four clean test tubes. The test tubes shall be placed in a water bath maintained at 212° ± 2°F., and melted wax added slowly to each tube until the strips are covered. A vented stopper shall then be used to close each test tube. After remaining in the water bath for 3 hours, the metal strips shall be removed from the test tubes, the wax completely wiped off the strips, and the strips examined visually for discoloration and corrosion. Slight surface etching shall not be considered as evidence of corrosion.

4.4.5 *Needle penetration.* The penetration of the wax shall be determined by method 214.0 of Specification SS-R-406, using a load of 100 grams for 5 seconds at 77°F.

4.4.6 *Acid number.* A sample of approximately 5 grams of the wax shall be carefully weighed, transferred to an Erlenmeyer flask, and 100 ml of neutral alcohol added. The flask shall be fitted with a reflux air condenser and the flask and its contents heated on a steam bath until the sample is melted. The mixture shall be agitated, cooled, and titrated with N/10 sodium hydroxide, using phenolphthalein as indicator. Calculate the acid number (milligrams of potassium hydroxide per gram of sample) as follows:

$$\text{Acid number} = \frac{56.1\ AN}{W}$$

where:

A = ml of sodium hydroxide solution used

N = normality of the sodium hydroxide solution used

W = weight of sample, in grams

4.4.7 *Fungistatic effectiveness.* The fungistatic effectiveness of the wax shall be determined following the procedure prescribed hereinafter.

4.4.7.1 *Preparation of cultures.*

4.4.7.1.1 *Test organisms.* The following fungi shall be used for testing the fungistatic effectiveness of the wax (see 6.2):

Name of organism	*American Type Culture Collection Number*
Aspergillus niger	9642
Aspergillus flavus	9643
Penicillium luteum	9644
Trichoderma T-1	9645

4.4.7.1.2 *Maintaining stock cultures.* Cultures of the fungi listed in 4.4.7.1.1 shall be maintained separately on an appropriate medium, such as potato-dextrose agar. The stock cultures shall be kept for not more than four months in a refrigerator at a temperature of from 3°C. to 10°C. Subcultures, incubated at 28°C. to 30°C. for 7 to 20 days, shall be used in preparing the spore suspension.

4.4.7.1.3 *Spore suspension.* The spore suspension shall be prepared by pouring into the subculture of each of the four fungi (see 4.4.7.1.1 and 4.4.7.1.2) a sterile 10-ml portion of a solution containing 0.05 grams per liter of a non-toxic wetting agent, such as dioctyl sodium sulfosuccinate (see 6.3), or approved equal, and shaking the container in order to get the spores into suspension. The spore suspensions of the individual fungi shall then be poured together and mixed to provide a composite spore suspension for use in the test. The spore suspension may be prepared fresh each day, or may be held in a refrigerator at a temperature of 3°C. to 10°C. for not more than 7 days.

4.4.7.1.4 *Test medium.* The test medium shall be made up and autoclaved, approximately 100 ml to a flask, and then poured into petri dishes with approximately 25 ml of the medium placed in each dish. The test medium shall have the following composition:

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Dipotassium hydrogen phosphate (K_2HPO_4).	0.7 grams
Potassium dihydrogen phosphate (KH_2PO_4).	0.7 grams
Magnesium sulfate ($MgSO_4 \cdot 7H_2O$)	0.7 grams
Ammonium nitrate (NH_4NO_3)	1.0 grams
Agar	20.0 grams
Distilled water	1.0 liter

4.4.7.2 *Preparation of test specimens.*

4.4.7.2.1 *Conditioning of wax.* Place approximately 300 ml of wax in a wide-mouthed beaker or jar, of approximately 400-ml capacity, and having a loosely-fitting cover. Heat the container and the wax in a well ventilated bath or oven at 105° ± 2°C. for 6 hours. Remove the container and the wax from the bath or oven and place them on a hot plate, keeping the wax at 100° ± 5°C.

4.4.7.2.2 *Test specimens.* A sheet of Whatman No. 2 filter paper, or approved equal, shall be dipped into the molten wax (see 4.4.7.2.1). The impregnated filter paper shall be withdrawn, the excess wax drained off, and cooled to room temperature. A total of 4 test specimens shall be cut or punched from the wax-coated filter paper for each wax sample tested. Each specimen shall have a diameter of 3 centimeters.

4.4.7.3 *Inoculation and incubation.* Each of the 4 test specimens, prepared as in 4.4.7.2.2, shall be layed on the center of the set agar in a separate petri dish (see 4.4.7.1.4). The surface of the medium as well as that of the test specimen shall then be inoculated with the composite spore suspension (see 4.4.7.1.3) either by spraying the suspension from a sterilized atomizer so that the entire surface is moistened by the suspension, or by delivering 0.5 to 1.0 ml of the suspension from a sterilized pipette, and tilting the dish from side to side in order to moisten the entire surface of the medium and the specimen. With each daily group of tests a piece of untreated filter paper, 3 centimeters in diameter, shall be placed on each of the 4 dishes of the set agar. These shall be seeded with the spore suspension, in the manner indicated above, to serve as controls.

4.4.7.4 *Evaluation of results.*

4.4.7.4.1 *Fungus growth in control dishes.* There shall be copious growth of fungus in each of the 4 control dishes. Absence of such growth requires repetition of the test, and shall not be construed as constituting a failure on the part of the wax.

4.4.7.4.2 *Fungus growth in test dishes.* The growth of the test fungi, as observed with the naked eye, shall not extend more than 2 millimeters from the edge toward the center of any test specimen. If only one of the test specimens show fungus growth exceeding the 2-millimeter limit the test may be repeated, in which case the wax shall be considered fungistatic only if all the test specimens meet the test requirement on the retest.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level C.* Unless otherwise specified, the wax shall be packed in standard commercial containers, or in wrapped packages so constructed as to protect the wax from contamination by foreign material during shipment and storage. The wax shall be packed in such a manner that it can be easily removed from the container or wrapper, and broken or cut into smaller pieces for easier melting prior to use.

5.2 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Cultures of the fungi mentioned in 4.4.7.1.1 may be obtained from the following sources:

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1. American Type Culture Collection
   2029 M Street, N. W.
   Washington, D. C.

2. Commanding General
   Wright Air Development Center
   ATTN: WCRTH-2
   Wright-Patterson Air Force Base
   Ohio

3. Quartermaster Research and
   Development Laboratories
   Philadelphia Quartermaster Depot
   2800 South 20th Street
   Philadelphia 45, Pa.

6.3 Dioctyl sodium sulfosuccinate is available commercially under the trade name of Aerosol OT.

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Preparing activity:
Army—Ordnance Corps

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ordnance
Air Force

Other interest:
Army—C
Navy—S

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1426

REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE.

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